Modeling Scaling of SWRO Membranes in Simscape

Nate DeGoede

1 The Motivation

Scaling in SWRO is the growth of mineral deposits on the surface of the membrane, reducing the effective membrane area and thus reducing the permeate production. What makes this a particularly compelling addition to the existing transient desalination modeling capabilities is that scaling is a transient phenomenon, and necessitates non constant operating conditions to be properly modeled.

2 Will it scale?

Lots of literature discusses predicting if scaling will happen or not. There are two main variables (very similar) that are used to predict scaling. The Saturation Index (SI) and the Supersaturation Ratio (S_r). Both are functions of the ion activity product (IAP) and the solubility product (K_{sp}). The equations are shown below:

$$SI = log\left(\frac{IAP}{K_{sp}}\right)$$

$$S_r = \sqrt{\frac{IAP}{K_{sp}}}$$

If $SI > 0$ or $S_r > 1$, scaling is likely to occur. Effectively these say the same thing if the ion activity product exceeds the solubility product, scaling is likely to occur.

The ion activity product is a function of the concentration of ions in solution, and can be calculated as follows:

$$IAP = \prod_{i=\text{ions}} \gamma_i[i]$$

where γ_i is the activity coefficient of ion i, and $[i]$ is the concentration of ion i in mol/L. We already have the concentration of ions in kg/m^3 from our solution domain, so we can convert to mol/L with an added fluid property for molar mass (M_x). The activity coefficient and the solubility product can probably be dependent variables in the future, but for now I will treat them as constant fluid parameters.

3 From looking at plots [1] and Nate thinking...

Just knowing if scaling will occur is not sufficient for what we would like to model. This does not help us track performance over time as scaling builds up, or help us understand operational strategies to mitigate scaling. To do this we need to model the rate of scaling. While there are many models for determining if scaling will occur, there are less models for scaling rate. From some plots online, it appears that the growth rate is an exponential function. The scale size vs time plots generally start low, then gow into exponential till it hits a max slope. The growth rate seems to be dependent on 2 things, concentration (or rather SI) and size of existing scale.

From a source, the growth rate appears to take this form:

$$\frac{dM_{\text{scale}}}{dt} = A(1 - e^{-t/\tau})$$

where A is a maximum growth rate, that we see once the scale has some sort of critical mass and τ is a time constant that determines how quickly we reach that max growth rate. This equation makes a lot of sense to me, as growth rate should increase as the scale gets larger (increased area for more particles to attach to), but at some point when the membrane is covered, the growth rate should plateau. But then it should probably also eventually reach a maximum scale mass. So maybe a better model would be:

$$\frac{dM_{\text{scale}}}{dt} = A(1 - e^{-t/\tau})(1 - \frac{M_{\text{scale}}}{M_{\text{max}}})$$

where M_{max} is the maximum mass of scale that can build up on the membrane. This equation captures the initial increase in growth rate as scale builds up, but also captures the eventual plateauing of scale mass as it approaches a maximum value.

Unfortunately, I have not been able to find any sources that use such an equation, most are far too detailed for our purposes. Moving forward I'm going to look into general crystal growth models instead of specifically SWRO scaling models.

4 From Textbook [2]

Rates of crystallization are typically empirical. There are two main stages. The diffusional stage is where solute is transported from the fluid and onto the crystal surface. The deposition stage is where the solutes on the crystal surface integrate into the lattice structure. The textbook describes both stages of growth with:

$$\frac{dm}{dt} = k_d A(c - c_i) = k_r A(c_i - c^*)^i$$

where k_d and k_r are the diffusion and deposition rate or reaction mass transfer coefficient constant, A crystal surface area, and c, c_i, and c^* are the solute concentrations in the bulk solution, at the interface and at equilibrium saturation.

Because c_i is a function of the mass transfer onto the crystal surface, the two equations can be linked together. If we assume the solvent does not experience any net mass transfer, c_i is governed by this differential equation:

$$\frac{dc_i}{dt} = \frac{dm_{\text{diffusion}}}{dt} - \frac{dm_{\text{deposition}}}{dt} = k_d A(c - c_i) - k_r A(c_i - c^*)^i$$

Since c_i can't easily be measured, it is common to simplify to this:

$$\frac{dm}{dt} = K_G(c - c^*)^s$$

In aqueous solutions, s is often between 1 and 2. K_G is related to k_d, k_r, s, and i, but it's probably better to just use K_G since again knowing c_i is difficult.

Interesting here that they do not use the saturation index or supersaturation ratio, but instead use the concentration directly.

5 Proposed Mass Model

I think we can use the textbook model to create a good A. The text book admits after it gives that model though that it is very simple and doesn't capture the size dependence of growth rate. By including that exponential term I had before I think we can capture that size dependence with a simple and intuitive time constant.

$$\frac{dM_{\text{scale}}}{dt} = K_G(c - c^*)^s(1 - e^{-t/\tau})$$

Often this equation ends up looking like one of the earlier equations depending on which stage is rate limiting. If diffusion is rate limiting, then $K_G{=}k_d A$ and $s{=}1$. If deposition is rate limiting, then $K_G{=}k_r A$ and $s{=}i$.

But I do still feel the lack of a maximum scale mass is a major gap.

6 Problems

As I implement this model I see some problems. First, there is no behavior to keep scale mass from dropping below zero. However, I think an improved model where the $e^{-t/\tau}$ term is replaced with a size dependent term.

The second problem is we need a new mass balance equation for the solute. Initially I was building the scaling model into the membrane eqs block. However, after thinking some more, I think a better approach is to have a separate scaling eqs block. The mass transfer of the scaling block can be summed with the mass transfer of the membrane block, and the properties of the volume will be routed to both blocks. This not only makes the mass balance work out, but also allows for modeling scaling in pipes generally, so not just on RO membrane. Note that we will need to make area an input rather than a parameter in the membrane EQs block in this version.

The final problem is one I don't have a good solution for yet. We can easily* create a model for membrane area loss as a function of scale mass. However, when I had been thinking about this problem before, I was only thinking of situations where there is only scaling on one side of the membrane. Without knowing which side of the membrane the scaling is on, if we assume scaling on one side is minor, an easy way to deal with this is just to use max(area loss A, area loss B), or to keep things smooth, area loss area loss A + area loss B. However, if we have significant scaling on both sides of the membrane, this approach will over predict area loss. There is likely a stats solution here.

7 Scale size dependence model

I want to replace this part of the equation:

$$(1 - e^{-t/\tau})$$

with something that depends on scale size. A sigmoid function would be a good choice here. Something like this:

$$\frac{1}{1 + e^{-A(M_{\text{scale}} - B)}}$$

Trick is what are A and B. A is similar to our time constant from before. B is critical because we want this function to be near 0 when M_{scale} is near 0. But the number makes that work is dependent on A.

Alternatively, since setting smooth limits is actually already a function in simscape, we can just use scale area, A. Just throwing area into the equation makes the equation look more like those first equations from the textbook:

$$\frac{dM_{\text{scale}}}{dt} = k_g A (c - c^*)^s$$

where k_g switches between k_d and k_r depending on the limiting mass transfer mode.

For now, I make A a linear function of M_{scale} using a "specific area" term, a [m^2/kg].

$$A = aM_{\text{scale}}$$

However, there also needs to be some limits. First, the scale area can't exceed a maximum area (area of membrane or pipe surface). Second, if the mass of the scale is zero, this means there would never be growth. To prevent this, we set the minimum area to the nucleation area. These limits are set using a Simscape blend function for each limit.

8 Scale Mass Limits

This is the trickier part, I'm ignoring max for now, but probably should add eventually.

The minimum is zero

9 Scale detachment

In backwashing, SWRO membranes don't just have the scale dissolved, but the scale is broken off of the membrane wall. Often using osmotic pressure as the driver. So for our scaling eqs block, we need an additional input of pressure difference, which represents the total pressure difference across the scale (both osmotic and hydraulic).

From here we need a model for rate of scale removal. In real systems, different spots on the scale will be bonded to the membrane with different strengths. For this simple model, I propose having one average bond strength term. The challenge is then modeling as a rate, when our parameter description of the system suggests the scale should just detach in one instant.

One paper I've reviewed focused on the mixing of the broken scale, so not very helpful here, but the lack of any mention of any scale breaking dynamics suggests and instead a strong focus on flushing suggests that this happens quickly.

Another paper [3] that seems more relevant for the type of modeling we do here looks at a mix of rinsing an backwashing, although not explicitly for SWRO, seems to have the relevant behavior described.

This paper models things in "reverse", calculating the pressure drop across the membrane during the cleaning process instead of using a pressure drop to drive the cleaning. They open by saying that the pressure required to clean at any time is simply the pressure required to clean at time 0, multiplied by two factors, one that represents the change in resistance, and one that represents the change in area. The area is what I want to focus on. It is unfortunate that they focus on area and not mass, but oh well, I can make do.

For the area change they use this equation:

$$\frac{dA_{cleaned}}{dt} = \alpha Q_{backwash}$$

where α [m^2/m^3] is doing a lot of work describing this process. The trick is how do I get α. They have some examples measured from experiments, but I kinda disagree with constants being here.

As more of the membrane is unblocked, it is going to required more flow rate to remove the same area of scale. Simply because initially all flow has to go though the scale, but as the scale is removed, much of the flow will avoid the scale.

10 Scale Growth - Parameters and Validation

From figure 3 in McPherson et al. (2022) [1], the max growth rate is 1.25×10^{-9} mol/cm^2/s. This is for $CaCO_3$, which has a molar mass of ~100 g/mol, making the max growth rate in kg 1.25×10^{-6}kg/m^2/s. Given our scale growth equation:

$$\frac{dM_{\text{scale}}}{dt} = k_g A(c - c^*)^s$$

we therefore need $k_g(c - c^*)^s$ to equal 1.25×10^{-6}kg/m^2/s. This will inform our choice of k_g. I'm going to stick with the s=1 assumption, and the paper give us the concentration used, Ca^{2+} is 20 mM and CO_3^{2-} is 10 mM. I find it rather annoying that they do't use the same for the two ions because this makes it require multi species. I'm going to just set the CO_3^{2-} concentration as my concentration since that is the limiting solute. 10 mM, given the $CaCO_3$ molar mass of ~100 g/mol and the H_2O molar mass of ~18 g/mol, is equivalent to 55.56 m(kg/kg). And if we assume a water density of ~998 kg/m^3, is equivalent to 55.4 kg/m^3, kilograms of solute per meter cubed of solvent. This is our c. Next we need the equilibrium concentration c^*. The McPherson paper lists the solubility product for calcite is 3.47 $\times$ 10^{-6} mol^2/L^2, corresponding to an equilibrium concentration of 5.89 $\times$ 10^{-5} mol/L or 5.89 $\times$ 10^{-3} kg/m^3. Given all that, to make the max slope appear, we need k_g to be:

$$k_g = \frac{1.25 \times 10^{-6} kgm^{-2}s^{-1}}{(55.4kg/m^3 - 5.89 \times 10^{-3}kg/m^3)} = 22.6 \times 10^{-9} m/s$$

Next parameter we need is the "specific area of scale", a. Since this effectively sets the speed at which we get to that max growth rate, we should relate it to the coefficient in front of the cubic term (5.76 $\times$ 10^{-16} mol cm^{-2} s^{-3}) from the McPherson plot. Following the same unit conversion steps for the linear coefficient, this is equivalent to 5.76 $\times$ 10^{-13} kg m^{-2} s^{-3}. Now comes the hard part. We don't have a term dependent on time cubed, but we do have an exponential that we would like to relate to this term. Rewriting our growth equation with a max slope, K_{max} [kg m^{-2} s^{-1}], and substituting in aM_{scale} in for A, we get

$$\frac{dM_{\text{scale}}}{dt} = K_{max} a M_{\text{scale}}$$

a simple first order ODE, where $(K_{max}a)^{-1}$ [s] is our time constant τ. This makes our solution:

$$M_{\text{scale}} = C_0 e^{t/\tau}$$

or for a unit area:

$$M_{\text{scale}}/A_{max} = C_0 e^{t/\tau}$$

Now because we bound the area, we don't truly see this behavior, but for this a coefficient, we only care about the behavior before hitting that max slope point. The y intercept from the McPherson plot is 1.48 $\times$ 10^{-8} mol cm^{-2}, or 1.48 $\times$ 10^{-5} kg m^{-2}. So our goal is to find a pair of C_0 and τ that best makes this match up:

$$M_{\text{scale}}/A_{max} = C_0 e^{t/\tau} = 1.48 \times 10^{-5} + 5.76 \times 10^{-13} t$$

over the time spanning 0s to ~750s. We can then use that to set both a, from τ ($(K_{max}a)^{-1} = \tau$), and $A_{\text{nucleation}}$, from C_0 ($\frac{A_{\text{nucleation}}}{aA_{max}} = C_0$).

Using MATLAB curve fitter, we get a value of 8.831×10^{-6} kg/m^2 for C_0 and 0.004545 s^{-1} for τ^{-1}. This fit is shown in the figure below.

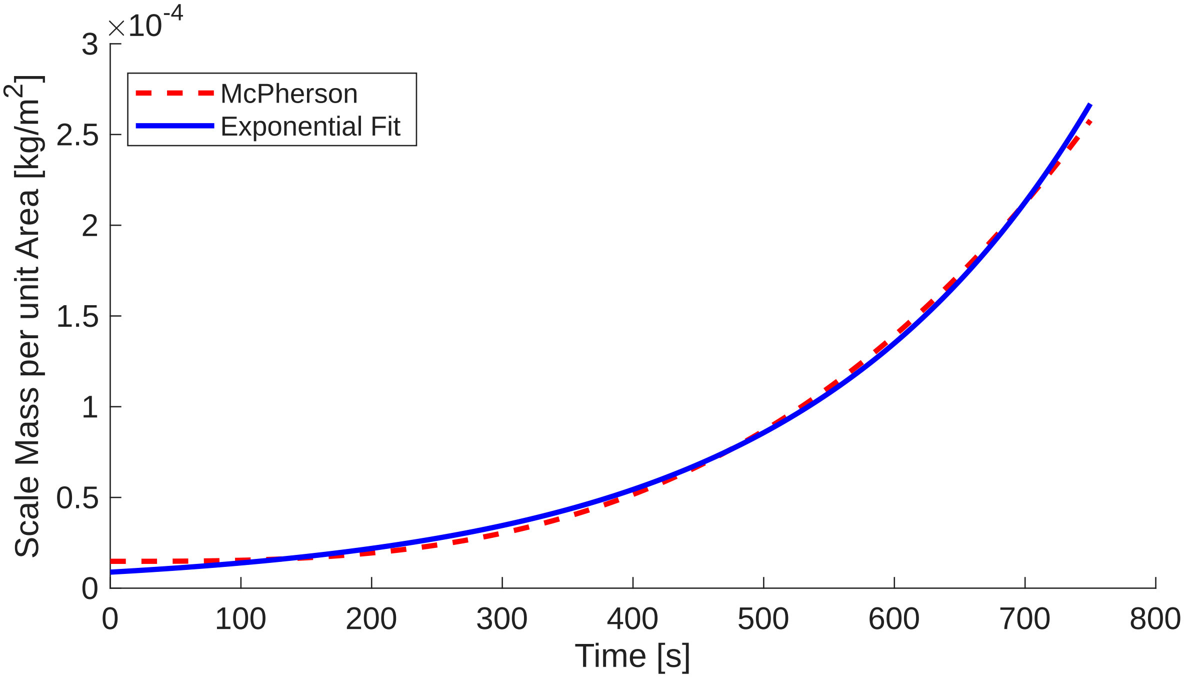

Figure 1: Scaling curve fit

These curve fit terms correspond to a specific area of 3636 m^2/kg and a nucleation area of 0.0321 A_{max}. These jump of the page as being a bit large to me, as 3% of the membrane covered at nucleation seems a bit much. The large specific area is responsible for both looking large. Although it is worth noting that this experiment is at a much smaller scale. Even so the 3636 m^2/kg seems really big for the specific area of the scale. Let's make a plot first, my goal is to replicate the figure 3 plot from McPherson using my new coefficients and solution domain model of scaling. McPherson uses a flow rate of 2 mL/min, in this case the pressure and temperature are not terribly important. This plots looks good though:

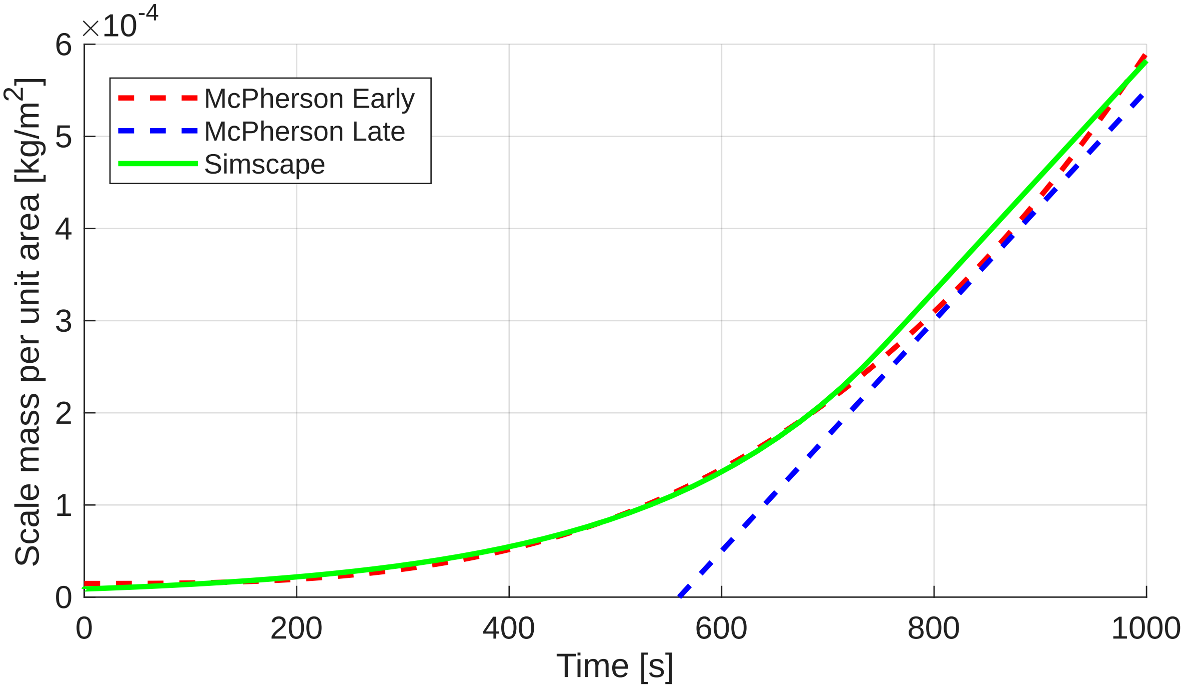

Figure 2: Simscape model vs. McPherson curves

You might wonder why there is a gap between the blue an green above. The reason for it is that at the larger scale masses, more of the solute concentration is on the scale instead of dissolved reducing the $(c - c^*)^s$ coefficient on the max growth rate. If I was tuning my model more I would use the concentration we see at the max scale growth rate point instead of the input concentration for c in the k_g tuning, and iterate this process until convergence.

11 Scale Detachment - Parameters and Validation

There are is a lot less to reference here. Many papers look only show before and after behavior, skipping dynamics during the cleaning processes, the thing I actually want to validate. One approach would be to just look at the flux recovery achieved and the time cleaned and then find an α that replicates the same flux recovery over the same time period. However, in many of these studies, like the 2005 paper from Sagiv and Semiat [4], the backwash procedure leaves the membrane completely cleaned, so there is no way of knowing if the full backwashing time used was necessary, or just used to ensure fully cleaned. Many of the studies that do actually look at the dynamics of the cleaning process are mostly looking at the mixing of the scale [5] [6], something that our Simscape model is not attempting to replicate. A more promising paper for validation is the 2019 paper by

Jepsen et al. Online Backwash Optimization of Membrane Filtration for Produced Water Treatment [7]. Interestingly this paper is actually doing something similar to what I'd like to do once this tool is fully built, but with hardware in the loop and for the removal of oil from water rather than salt. Despite the differences in application, I think this is actually a good demonstration of the diversity of the tool. Although it will not solve the issue of me needing parameters for my SWRO study.

From that Jepsen paper, I'll look at the cleaning stage between 2000 and 4000 seconds to find parameters. In that cleaning stage, the backwash flux is ~570 L/(m^{2}hr) for 600s. And the permeate resistance goes from 3.00×10^{-3} to 2.03×10^{-3} bar hr m^2/L. This resistance accounts for the the loss in area. Our modified membrane flux equation is now:

$$A_w A_m (1 - a_{loss})\big((P_A - P_B) - (\pi_A - \pi_B)\big) = \frac{\dot{m}_{w,B}}{\rho_w(T, P_B)}$$

where a_{loss} is the fraction of area covered by the scale/fouling. So in this study, the "flux resistance" is:

$$R = \big[A_w(1 - a_{loss})\rho_w(T, P_B)\big]^{-1}$$

Assuming A_w and $\rho_w(T, P_B)$ remain constant, that means the change in resistance we saw earlier is entirely due to the a_{loss}.

$$\frac{R_1}{R_2} = \frac{A_w(1 - a_{loss,2})\rho_w(T, P_B)}{A_w(1 - a_{loss,1})\rho_w(T, P_B)} = \frac{1 - a_{loss,2}}{1 - a_{loss,1}}$$

Additionally, by looking at the resistance at the start, I'll assume that is the case for $a_{loss} = 0$. The resistance here is 0.85×10^{-3} bar hr m^2/L. So I also have:

$$\frac{R_1}{R_0} = \frac{1}{1 - a_{loss,1}}$$

Giving me enough info to solve for the a_{loss} values. The total area recovery in terms of a_{loss} is

$$A_m(a_{loss,2} - a_{loss,2}) = Q_b \alpha$$

where Q_b is the backwashing flow rate. So now we have enough information to find α.

12 Scale Detachment - Parameters take 2

There's not a huge demand it seems to understand the dynamics during the backwash stage, just knowing roughly how long to clean is good enough. So I will use the 2005 paper from Sagiv and Semiat [4] to get a rough number. Fig 3, shows the impact of cleaning recovering 5% of the flux with a 20s backwash process. Fig 4, shows that for a 20s backwash, the volume of permeate used is roughly 95 ml (V_b). This give us enough information to solve for α, although it is worth noting that since it gets fully cleaned, it α could be higher than what we will calculate.

First step to calculate α is figuring out the a_{loss} coefficient:

$$Q_1 = A_w A_m \big((P_A - P_B) - (\pi_A - \pi_B)\big)$$
$$Q_2 = A_w A_m (1 - a_{loss}) \big((P_A - P_B) - (\pi_A - \pi_B)\big) = Q_1(1 - 0.05)$$
$$a_{loss} = 0.05$$

Then with this, α is calculated through:

$$V_b \alpha = a_{loss} A_m$$

The membrane area in this study is 0.48 m^2. So that makes the α 253 m^2/m^3.

An experiment that would give us a better understanding of the alpha parameter would have been if this paper had either stopped before fully cleaned, or even better, stopped at a variety of different times.

13 Scale impact on major losses

The build up of scale leads to increased head losses through a pipe (or past a membrane). This increase occurs due to two mechanisms, decreased hydraulic diameter (scale takes up space), and increased surface roughness (the scale is not as smooth as the pipe or membrane). Loss of diameter in circular pipes is easy enough, if we use the mass M, density ρ, and area of scale A, we can easily find the scale height h through the following

$$M = \rho A h$$

Since we have mass and area already and density could be a fluid property, finding h is easy. Then you'd just subtract this from the hydraulic diameter for cylindrical pipes. For other geometries, like with RO membranes, you'd need to do math, but since for RO membranes, it is not advised to run the system past the point of full coverage, or even approach that point.

Looking through literature for roughness impacts of scaling

14 Take 2 on roughness change due to scaling

Our total ϵ_{total} will be determined by:

$$\epsilon_{total} = (1 - a_{loss})\epsilon_{pipe} + a_{loss}\epsilon_{scale}$$

where ϵ_{pipe} and ϵ_{scale} are the roughnesses of the pipe and scale respectively. The pipe roughness is set, so the trick is figuring out the scale roughness. Given a density of scale ρ_s, mass M, and area Aa_{loss}, we can easily find the average height, h_{avg}.

$$h_{avg} = \frac{M}{\rho_s A a_{loss}}$$

From the above equation, we can substitute our specific area term a to create this equation

$$h_{avg} = \frac{1}{\rho_s a}$$

We will set the standard deviation of height, σ_h as a parameter, could eventually become a function of mass. Then we use this equation to define the ϵ_{scale}

$$\epsilon_{scale} = h_{avg} + \beta\sigma_h$$

Alternatively, using a coefficient of variation S instead of σ_h

$$\epsilon_{scale} = h_{avg}(1 + \beta S)$$

15 Other papers

[8] [9] [10] [11] [12] [13] [14] [15]

[1] McPherson, I. J., Peruffo, M., and Unwin, P. R., 2022, "Role of Mass Transport in the Deposition, Growth, and Transformation of Calcium Carbonate on Surfaces at High Supersaturation," Crystal Growth & Design, **22**(8), pp. 4721–4729. https://doi.org/10.1021/acs.cgd.1c01505.

[2] RICHARDSON, J. F., HARKER, J. H., and BACKHURST, J. R., 2002, "Crystallisation," *Chemical Engineering*, Elsevier, pp. 827–900. https://doi.org/10.1016/b978-0-08-049064-9.50026-6.

[3] Guo, J., and Wang, Z., 2024, "A New Mathematical Model to Describe the Trans-Membrane Pressure in the Coupled Cleaning Process (Rinsing + Backwash)," Separation and Purification Technology, **337**, p. 126354. https://doi.org/10.1016/j.seppur.2024.126354.

[4] Sagiv, A., and Semiat, R., 2005, "Backwash of RO Spiral Wound Membranes," Desalination, **179**(1), pp. 1–9. https://doi.org/https://doi.org/10.1016/j.desal.2004.11.050.

[5] Ramon, G., Agnon, Y., and Dosoretz, C., 2010, "Dynamics of an Osmotic Backwash Cycle," Journal of Membrane Science, **364**(1-2), pp. 157–166. https://doi.org/10.1016/j.memsci.2010.08.008.

[6] Sagiv, A., and Semiat, R., 2010, "Modeling of Backwash Cleaning Methods for RO Membranes," Desalination, **261**(3), pp. 338–346. https://doi.org/10.1016/j.desal.2010.04.054.

[7] Jepsen, K. L., Bram, M. V., Hansen, L., Yang, Z., and Lauridsen, S. M. Ø., 2019, "Online Backwash Optimization of Membrane Filtration for Produced Water Treatment," Membranes, **9**(6), p. 68. https://doi.org/10.3390/membranes9060068.

[8] Darkins, R., Kim, Y.-Y., Green, D., Broad, A., Duffy, D., Meldrum, F., and Ford, I., 2022, "Calcite Kinetics for Spiral Growth and Two-Dimensional Nucleation," Crystal Growth & Design, **22**. https://doi.org/10.1021/acs.cgd.2c00378.

[9] Qin, J.-J., Oo, M. H., Kekre, K. A., and Liberman, B., 2010, "Development of Novel Backwash Cleaning Technique for Reverse Osmosis in Reclamation of Secondary Effluent," Journal of Membrane Science, **346**(1), pp. 8–14. https://doi.org/https://doi.org/10.1016/j.memsci.2009.08.011.

[10] Kim, M., Park, J., Park, K., Kim, P., Kim, H., and Kim, J., 2020, "Evaluation of Forward Osmosis Backwash in a Pilot-Scale Seawater Reverse Osmosis (SWRO) Desalination System," Desalination and Water Treatment, **173**, pp. 1–11. https://doi.org/https://doi.org/10.5004/dwt.2020.24696.

[11] Ahmed, M. A., Amin, S., and Mohamed, A. A., 2023, "Fouling in Reverse Osmosis Membranes: Monitoring, Characterization, Mitigation Strategies and Future Directions," Heliyon, **9**(4), p. e14908. https://doi.org/10.1016/j.heliyon.2023.e14908.

[12] Sagiv, A., and Semiat, R., 2010, “Parameters Affecting Backwash Variables of RO Membranes,” Desalination, **261**(3), pp. 347–353. https://doi.org/10.1016/j.desal.2010.04.012.

[13] Li, C., Ning, Y., Yan, T., and Zheng, W., 2019, “Studies on Nucleation and Crystal Growth Kinetics of Ferrous Oxalate,” Heliyon, **5**(11), p. e02758. https://doi.org/10.1016/j.heliyon.2019.e02758.

[14] Løge, I. A., and Anabaraonye, B. U., 2024, “The Importance of Detachment Processes in Modeling Crystallization Fouling,” Chemical Engineering Journal, **497**(Denmark), p. 154642. https://doi.org/10.1016/j.cej.2024.154642.

[15] Sagiv, A., Semiat, R., and Shemer, H., 2023, “Simulation of CaCO3 Reverse Osmosis Membrane Scaling at a Continuously Increasing Scaling Propensity,” Desalination and Water Treatment, **294**, pp. 20–28. https://doi.org/https://doi.org/10.5004/dwt.2023.29595.